

Literature Review on Graph Neural Networks for Fault Detection and Localization in Power Grids

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June 2026

Abstract

Fault detection and localization are essential for secure and reliable smart-grid operation. As power systems integrate renewable energy sources, bidirectional power flows, distributed generation, and high-frequency sensing devices such as phasor measurement units (PMUs), conventional protection and monitoring methods face increasing difficulty. This literature review examines major research directions for power-system fault detection and localization, including traditional relay-based methods, shallow machine learning, deep learning, graph neural networks (GNNs), and spatiotemporal learning approaches. The review shows that topology-agnostic methods often fail to represent the physical structure of power networks, while static graph methods may underuse the temporal evolution of fault transients. Spatiotemporal GNNs provide a promising direction because they combine graph-based spatial learning with temporal sequence modeling. The review concludes by identifying key research gaps related to bus-level localization, robustness, real-data validation, and multi-task learning.

Keywords: literature review, graph neural networks, smart grid, fault detection, fault localization, PMU, GCN, LSTM, spatiotemporal learning.

1 Introduction

Modern power grids are becoming more complex because of renewable energy integration, changing load behavior, distributed energy resources, and high-speed measurement systems.

These changes make fault detection and localization more challenging. A protection system must not only detect whether a fault has occurred, but also identify where the fault originated so that corrective action can be taken quickly.

The literature on this problem has evolved through several stages. Early approaches used impedance-based relays, threshold logic, and signal-processing techniques. Later, researchers introduced shallow machine-learning models such as support vector machines (SVMs), random forests, and artificial neural networks (ANNs). More recent studies use deep learning, including convolutional neural networks (CNNs), recurrent neural networks, and GNNs.

The central challenge is that power-system faults are both spatial and temporal. They are spatial because disturbances propagate through buses, lines, transformers, and electrical coupling. They are temporal because voltage, current, frequency, and power signals evolve rapidly during pre-fault, fault, and post-fault periods. A strong fault-localization model should therefore represent both the grid topology and the transient signal behavior.

This review presents the main research directions and explains why spatiotemporal GNNs are a strong candidate for real-time fault detection and localization in power grids.

2 Traditional Fault Detection and Localization

Traditional fault detection methods include impedance-based relaying, overcurrent protection, differential protection, traveling-wave analysis, and threshold-based monitoring. These methods are widely used because they are fast, interpretable, and suitable for real-time protection hardware. Their decisions are also easier for system operators to understand because they are based on well-known electrical principles.

However, traditional methods depend heavily on accurate system parameters, relay settings, line impedance values, and stable operating assumptions. In modern grids, these assumptions are less reliable. Renewable generation, inverter-based resources, topology changes, and bidirectional power flow can alter fault signatures and reduce the reliability of fixed rules.

Liao et al. [1] discuss how smart-grid operation requires methods that can process large volumes of measurement data and adapt to changing system behavior. Chen et al. [3] also emphasize that data-driven fault diagnosis can address limitations of purely rule-based systems. Traditional methods remain important, but they are often insufficient for flexible, high-dimensional, PMU-based fault localization.

3 Shallow Machine-Learning Approaches

Shallow machine-learning methods were introduced to improve adaptability. Common examples include SVMs, decision trees, random forests, k-nearest neighbors, and shallow ANNs. These methods can learn nonlinear decision boundaries from measured voltage, current, power, or frequency features.

The main advantage of shallow learning is that it can outperform simple threshold rules when the training data are representative. For example, a classifier can learn patterns associated with different fault types or approximate fault locations. These models are also relatively easy to train compared with large neural networks.

The major limitation is feature dependence. Shallow models usually require manually designed features such as RMS values, harmonic components, sequence components, wavelet features, or statistical summaries. If the operating condition changes, those handcrafted features may no longer separate fault classes reliably. In addition, shallow models usually treat bus measurements as ordinary feature vectors and do not explicitly represent the physical grid topology.

For bus-level fault localization, ignoring topology is a serious limitation. Two buses may be electrically close even if their feature columns are far apart in a vector representation. This motivates graph-based learning methods that directly encode the power-network structure.

4 Deep Learning for Fault Diagnosis

Deep learning reduces the need for manual feature extraction. CNNs can learn patterns from waveform images, time-frequency maps, or bus-feature matrices. Recurrent models such as LSTMs can learn temporal patterns from sequential PMU windows.

CNN-based methods are useful because they can automatically detect local patterns in data. However, Owerko et al. [2] point out that power systems are not naturally arranged as Euclidean grids. In fault localization, this matters because CNNs depend on the ordering of input rows and columns. If bus ordering changes, the physical system remains the same, but the CNN input pattern changes.

LSTM networks, introduced by Hochreiter and Schmidhuber [8], are useful for sequential data. In power-system fault analysis, LSTMs can capture how signals evolve over short time windows. This is important because early transient behavior often contains useful information about fault type and location.

However, a pure LSTM over flattened bus measurements does not know which buses are connected. It can learn time dynamics, but it does not explicitly model grid topology. Therefore, deep learning methods still need a way to combine temporal learning with physical network structure.

5 Graph Neural Networks in Power Systems

GNNs are well suited for power systems because a grid can naturally be represented as a graph. Buses are nodes, transmission lines and transformers are edges, and measurements such as voltage magnitude, phase angle, active power, reactive power, and frequency deviation are node features.

Kipf and Welling [7] introduced a graph convolutional framework that updates node representations by aggregating information from neighboring nodes. In a power grid, this is meaningful because disturbances propagate through electrical connections. A GNN can therefore learn from physically connected buses rather than from arbitrary feature ordering.

Chen et al. [3] applied graph convolutional networks to power-system fault diagnosis and showed that topology-aware learning can improve diagnosis performance. Donon et al. [4] explored graph neural solvers for power-system problems and showed that graph learning can capture structural properties of electrical networks. James et al. [6] investigated GCNs for fault location in distribution networks, showing the relevance of graph learning for localization tasks.

The strength of GNNs is physical consistency. If the graph representation is constructed correctly, the model can learn from electrical neighborhoods instead of relying on arbitrary bus indexing. GNNs also scale well for sparse networks, since real power grids have far fewer lines than all possible bus-to-bus connections.

The limitation is that many GNN-based approaches use static snapshots. A single post-fault snapshot may not capture enough information to distinguish the true fault location, especially when neighboring buses experience similar disturbances. This leads naturally to spatiotemporal graph learning.

6 Spatiotemporal Learning

Spatiotemporal learning combines spatial modeling with temporal modeling. Yu et al. [5] proposed a spatiotemporal graph convolutional network for traffic forecasting, showing that graph structure and temporal dynamics can be learned jointly. Although traffic networks and power grids are different, both involve signals that evolve over time on a network.

For power-system fault localization, the same idea is highly relevant. A GCN can model how the fault disturbance spreads across connected buses, while an LSTM or temporal convolution can model how the disturbance evolves across a PMU window. This combination is especially useful for real-time localization because the early transient may contain the strongest location-specific information.

A hybrid GCN-LSTM structure is therefore a practical architecture for this problem.

The GCN component handles spatial propagation, and the LSTM component handles temporal progression. A multi-task objective can further improve learning by training the model to predict both fault location and fault type.

7 Comparison of Method Families

Table 1 summarizes the major method families reviewed in this document.

Table 1: Comparison of Method Families for Power-Grid Fault Analysis		
Method Family	Strength	Limitation
Relay and threshold methods	Fast, interpretable, practical for protection systems	Sensitive to parameter uncertainty and changing grid conditions
Shallow machine learning	Learns nonlinear patterns from measured data	Requires handcrafted features and often ignores topology
CNN-based deep learning	Learns features automatically from waveform or matrix inputs	Assumes Euclidean structure and depends on bus ordering
LSTM/recurrent models	Captures temporal evolution of fault signals	Does not explicitly model physical grid connectivity
GNN/GCN models	Uses grid topology directly through message passing	Often applied to static snapshots without temporal context
Spatiotemporal GNNs	Models both grid topology and time-varying transients	Requires careful dataset design and robustness validation

8 Research Gap

The reviewed literature reveals several research gaps. First, many studies emphasize fault detection or fault-type classification, while bus-level localization remains more difficult. Second, topology-agnostic models may perform well on fixed datasets but lack physical consistency under bus reordering or topology changes. Third, spatial GNN models often underuse temporal fault signatures. Fourth, many studies report accuracy but provide limited robustness analysis under measurement noise, class imbalance, and N-1 contingencies.

These gaps suggest the need for a model that jointly learns topology and time. A spatiotemporal GNN can address this by using graph convolution for spatial dependencies and sequence modeling for transient behavior. Multi-task learning can also help because fault type and fault location are related outputs.

9 Direction for Current Research

The reviewed works support a research direction based on three ideas. First, fault signals are graph-structured because physical connections determine how disturbances propagate. Second, fault signals are time-dependent because useful localization information appears during short transient windows. Third, fault location and fault type are related, so a shared representation can support both tasks.

For this reason, a model that combines GCN layers, LSTM layers, and multi-task output heads is well aligned with the limitations found in existing work. Synthetic data generation tools such as Pandapower [9] can support systematic experiments by varying bus location, fault type, fault resistance, and loading level.

10 Conclusion

This literature review shows a clear progression from traditional protection methods to topology-aware spatiotemporal learning. Traditional and shallow-learning methods remain useful, but they struggle with high-dimensional PMU data, changing grid conditions, and bus-level localization. CNN and LSTM methods improve feature learning but do not fully solve the topology-time coupling problem. GNNs provide a natural way to represent power grids, while spatiotemporal GNNs extend this advantage by modeling transient dynamics.

Overall, the literature supports the use of spatiotemporal graph neural networks for real-time fault detection and localization. Future research should focus on real-data validation, robustness under noise and topology changes, explainability, and transfer learning across different grid systems.

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